



Cost Optimization Solutions For E-Commerce Vendors

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Introduction

In today's times, retail is soaring with the advent of e-commerce and a majority of sellers are coming online to sell their products. This has resulted in an increased competition.

The main reason behind the selection of this topic is to provide not only the customers but also the vendors with the most cost-effective solution. In many cases, it is observed that the prices listed for products may not be the best ones.

This topic is relevant to the current world, with respect to the fact that majority of the population has shifted to online shopping. This is due to the fact that the market giants are using a lot of money to please the customers.

Since 2017, the percentage of online sales has risen from 34.5% to 53.9% in 2021, in the US alone. This trend suggests that there is not going to be a lowering in the percentages of online sales anytime soon.

Materials and methods

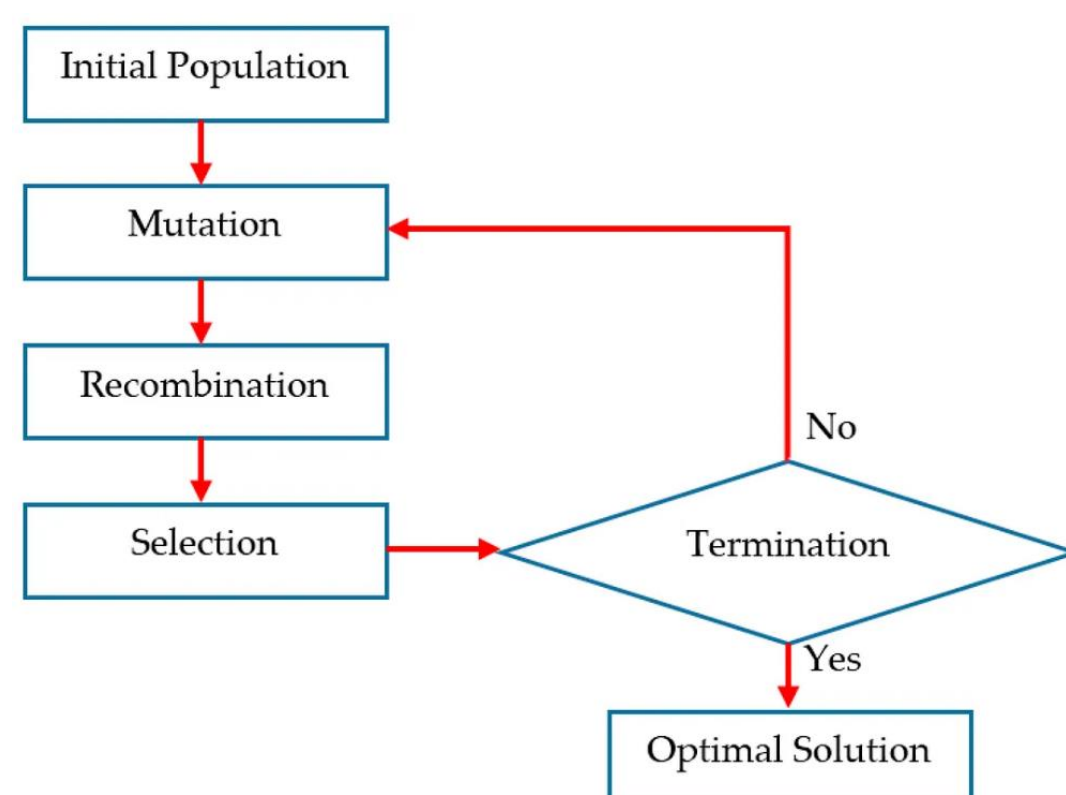


Figure 2. Flowchart depicting Differential Evolution

Differential evolution (DE) is a population-based metaheuristic search algorithm that optimizes a problem by iteratively improving a candidate solution based on an evolutionary process. Differential evolution can be considered to be one of the most robust algorithms based on population. It does not make any assumption about the underlying information and can explore the entire data very speedily.

The algorithm works by having a set of initial agents and creates the next set of agents by using a set of parameters. This next set of agents created is such that they have the best objective values.

If not, then they are simply discarded. This entire process keeps repeating itself until a certain criterion is not met. When this happens, the algorithm stops.

Mathematical Model

Material Cost

$$= \sum A_{v,m} \times C_{v,m} \forall v, m$$

where v = vendor, m = material

Production Cost

$$= \sum A_{g,f} \times C_{g,f} \forall f, g$$

where g = goods, f = facility

Transportation Cost

$$= \sum A_{m,v,f} \times C_{m,v,f} \forall v, f, m +$$

$$\sum A_{g,f,w} \times C_{g,f,w} \forall f, w, g +$$

$$\sum A_{g,w,c} \times C_{g,w,c} \forall w, c, g$$

Where v = vendor, f = facility, m = material, w = warehouse, g = goods, c = customer

Inventory Cost = IC

Objective Function =

$$\text{Min } (\sum A_{v,m} \times C_{v,m} \forall v, m + \sum A_{g,f} \times C_{g,f} \forall f, g + \sum A_{m,v,f} \times C_{m,v,f} \forall v, f, m + \sum A_{g,f,w} \times C_{g,f,w} \forall f, w, g + \sum A_{g,w,c} \times C_{g,w,c} \forall w, c, g + IC)$$

Limits Applied

- $0 \leq A_{m,v,f} \leq A_{m,v,f}(\text{max})$
- $0 \leq A_{g,f,w} \leq A_{g,f,w}(\text{max})$
- $0 \leq A_{g,w,c} \leq A_{g,w,c}(\text{max})$
- $A_{g,f}(\text{min}) \leq A_{g,f} \leq A_{g,f}(\text{max})$

Restrictions

- $A_{m,v,f} = (\text{Amount sold})_{v,f} \forall v, f, m$
- $A_{g,w,c} = (\text{Amount sold})_{g,c} \forall w, c, g$

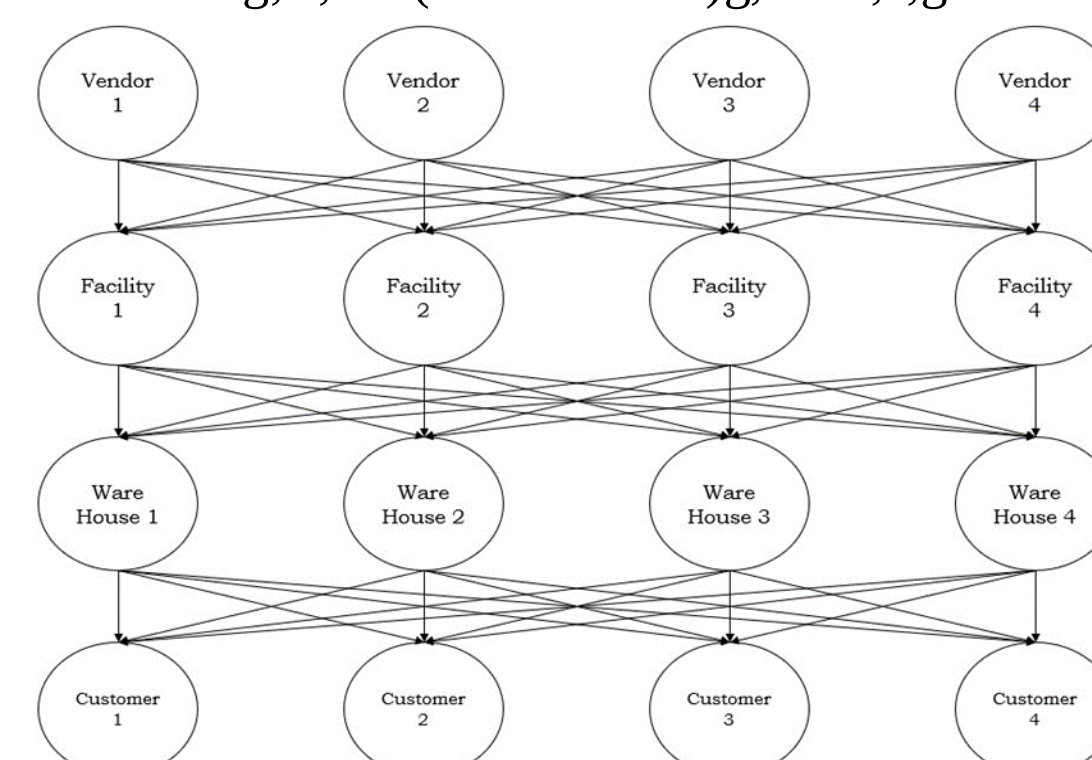


Figure 4. Network Diagram

Data Used

		MC	PC	TC			IC
				3 kms	51.5 kms	100 kms	
Inbound Delivery	Open	43.12	5.7 (milk + 1 (add-on))	0.04	0.67	1.3	150000000
	Tanker			0.03	0.47	0.91	
Outbound Delivery	Wholesale			0.05	0.9	1.75	
	Retailer			0.03	0.45	0.87	

Results

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Iteration: 7 f([[0.97841 0.96458]]) = 0.00580
Iteration: 12 f([[1.02934 1.05441]]) = 0.00350
Iteration: 14 f([[1.02959 1.05721]]) = 0.00168
Iteration: 27 f([[0.95945 0.92093]]) = 0.00166
Iteration: 29 f([[0.96362 0.92824]]) = 0.00133
Iteration: 33 f([[0.96453 0.93014]]) = 0.00126
Iteration: 34 f([[1.02237 1.04274]]) = 0.00112
Iteration: 35 f([[1.02259 1.04614]]) = 0.00053
Iteration: 36 f([[0.99409 0.98853]]) = 0.00004
Iteration: 45 f([[1.00314 1.00583]]) = 0.00003
Iteration: 47 f([[0.99619 0.99218]]) = 0.00002
Iteration: 48 f([[1.00028 1.00065]]) = 0.00000

Solution: f([[1.00028 1.00065]]) = 0.00000
  
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Figure 5. Optimization of Rosenbrock Function

Rosenbrock Function is given as:

$$F(\mathbf{x}) = \sum_{i=1}^{d-1} [100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]$$

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Solution: f([[ -1.45067 0.24073 ]]) = 0.03000
Process finished with exit code 0
  
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Figure 6. Optimization without Inventory Cost

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Solution: f([[ 1.60935 -3.22577 ]]) = 0.03000
Process finished with exit code 0
  
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Figure 7. Optimization with Inventory Cost

Conclusions

For this project, I have studied the Supply Chain Management and the effect of e-commerce on the retail sector. Along with this, a basic cost optimization model has been developed. The different costs have been identified and the objective function has been developed. The objective function is the minimization of the total cost. This minimization is to be achieved using an iterative algorithm.

I have used the very widely popular Differential Evolution for this. I have implemented the algorithm using the Rosenbrock Function first, before implementing it using the collected real world data.

In conclusion, such a solution strategy is a very viable option for such optimization problems.

Literature cited

- [1] S. Mansuri. "How retail and E-Commerce can scale fast using logistics optimization." peerbits. <https://www.peerbits.com/blog/retail-e-commerce-can-scale-fast-using-logistics-optimization.html>
- [2] K. Venkata Subbair, " Supply Chain Management in a Dairy Industry – A Case Study", Proceedings of the World Congress on Engineering 2009 Vol I WCE 2009, July 1 - 3, 2009, London, U.K
- [3] Georgioudakis, M. and Plevris, V. (2020) 'A comparative study of differential evolution variants in constrained structural optimization', Frontiers in Built Environment, 6. doi:10.3389/fbuil.2020.00102.
- [4] Lu, Lauren & Swaminathan, Jayashankar. (2015). Supply Chain Management. International Encyclopedia of the Social & Behavioral Sciences. 10.1016/B978-0-08-097086-8.73032-7.
- [5] Kleab, K. (2017) Important of supply chain management - IJSRP. Available at: <https://ijsrp.org/research-paper-0917/ijsrp-p6949.pdf>